

# Abdelrahman Abozena

Agentic AI Engineer | LLM & Backend Systems | Python

Giza, Egypt | boodyabozena@gmail.com | LinkedIn | GitHub

## SUMMARY

Targeting: Software Engineer: Internship Opportunities - Ignite Program for People with Disabilities at Microsoft

Agentic AI engineer and Computer Science student with professional experience leading AI initiatives and building backend AI

systems. Developed local-first coding agents, multi-agent simulation platforms, and event-driven AI pipelines using Python,

FastAPI, Ollama, Kafka, WebSockets, and retrieval/memory architectures.

## SKILLS

Agentic AI: multi-agent systems, context management, agent orchestration, tool calling, agent memory, LLM agents, LangGraph, RAG

Engineering & ML: model evaluation, React, Git, scikit-learn, TypeScript, GitHub, Docker, pytest, Vite

Backend & Data: Python, WebSockets, REST APIs, FastAPI, MongoDB, SQLite, Kafka, MySQL

Transformers, Ollama

## PROJECTS

Local-First Coding Agent

Python | Ollama | SQLite | AST/DOM Analysis | Multi-Agent Orchestration | Terminal UI

- Built a plan-first coding agent that improves small local models on long repository tasks through durable goals, approval-bound

plans, task DAGs, generated worker roles, and evidence-gated completion.

- Implemented repository indexing, hybrid lexical/graph/semantic retrieval, crash recovery, independent review, deterministic

verification, and code-enforced workspace safety.

Agentic AI Community Simulator

Python | FastAPI | MySQL | WebSockets | React | TypeScript | Multi-Agent Systems

- Built a full-stack platform where AI personas debate, influence one another, and surface product-validation and market signals

around user-defined ideas.

- Developed a FastAPI/MySQL backend with live WebSocket events and React workflows for research, Idea Court, Persona Lab,

authentication, administration, and simulation monitoring.

Real-Time Comment Intelligence Pipeline

Python | Kafka | FastAPI | Ollama | SQLite | React | Vite

- Built an event-driven pipeline across seven Kafka topics for sentiment analysis, coordinated-behavior detection, alerting, dead-

letter handling, and live monitoring.

- Connected Python workers, local Ollama inference, FastAPI APIs, SQLite state, and a React/Vite dashboard with health and

operational endpoints.

## Prompt Optimizer for Coding Agents

Python | FastAPI | React | Ollama | Git Diffs | Hybrid Retrieval | pytest

- Built a system that converts user intent, Git commits, code diffs, and unresolved ambiguity into implementation-ready prompts for

coding agents.

- Added persistent project memory, commit-gap tracking, hybrid retrieval, high-severity blind-spot gates, Ollama model selection,

and automated failure-path tests.

## EXPERIENCE

Foursw | Head of Artificial Intelligence | Jun 2026 - Present | Part-time | Hybrid | Giza, Egypt

- Lead the artificial intelligence function and coordinate the design and delivery of applied AI initiatives across product and

engineering workflows.

FlyRank AI | Backend AI Engineering Intern | May 2026 - Present | Remote | Giza, Egypt

- Contribute to backend AI engineering, focusing on server-side AI workflows, model-powered services, and system integration.

Instant Software Solutions | AI & Data Science Engineer | Jan 2026 - Present | On-site | Giza, Egypt

- Work across applied AI and data science projects involving machine learning, NLP, model evaluation, and backend AI systems.

EgenSolve | Backend Developer | Mar 2025 - Sep 2025 | Giza, Egypt

- Developed backend workflows and APIs for centralized employee account and issue management, connecting users with

developers to streamline problem resolution.

## EDUCATION

Egyptian Chinese University | B.Sc. in Computer Science | Expected 2027 - AI-focused coursework

- Harbin Institute of Technology - AI Summer School: Generative AI and Probabilistic Models (On-site, China)
- Digital Egypt Pioneers Initiative (DEPI) - Machine Learning Engineer Internship | Nov 2025 - Present
- Google x Kaggle - AI Agents Intensive Certificate